



Previous rollout design

$$N_{\text{strategy}} = PG$$

$$N_{\text{code}} = PG$$

Example: $P = 8$, $G = 64$ gives **512 strategies** and **512 programs**.

Current LORA policy: executable code rollouts, followed by verification and training

Hierarchical rollout design
 $s_{p,j} \sim \pi_{\text{strategy}}(\cdot \mid x_p, s_{p,<j})$ $y_{p,j,m} \sim \pi_{\theta}(\cdot \mid x_p, s_{p,j})$
 $P = 8, S = 4, C = 20$: $N_{\text{strategy}} = PS = 32$, $N_{\text{code}} = PSC = 640$.
- **16× fewer strategy generations** than the previous example.